

Angular Hands-On 7

Routing & Route Guards

Topics Covered

- Angular Router basics
- Configuring routes and navigation
- RouterLink and RouterOutlet
- Route Guards (canActivate)
- Protecting pages with authentication logic

Task 1: Configure Routing

Goal: Set up navigation between pages.

Steps:

1. In app-routing.module.ts, define routes for Home, Courses, Enroll, and Profile.
2. Add <router-outlet> in app.component.html.
3. Add navigation links using .

Expected Outcome: Clicking links navigates between pages without full reload.

Task 2: Implement Route Guard

Goal: Protect the Profile page with an authentication guard.

Steps:

1. Generate guard:

bash

ng generate guard guards/auth

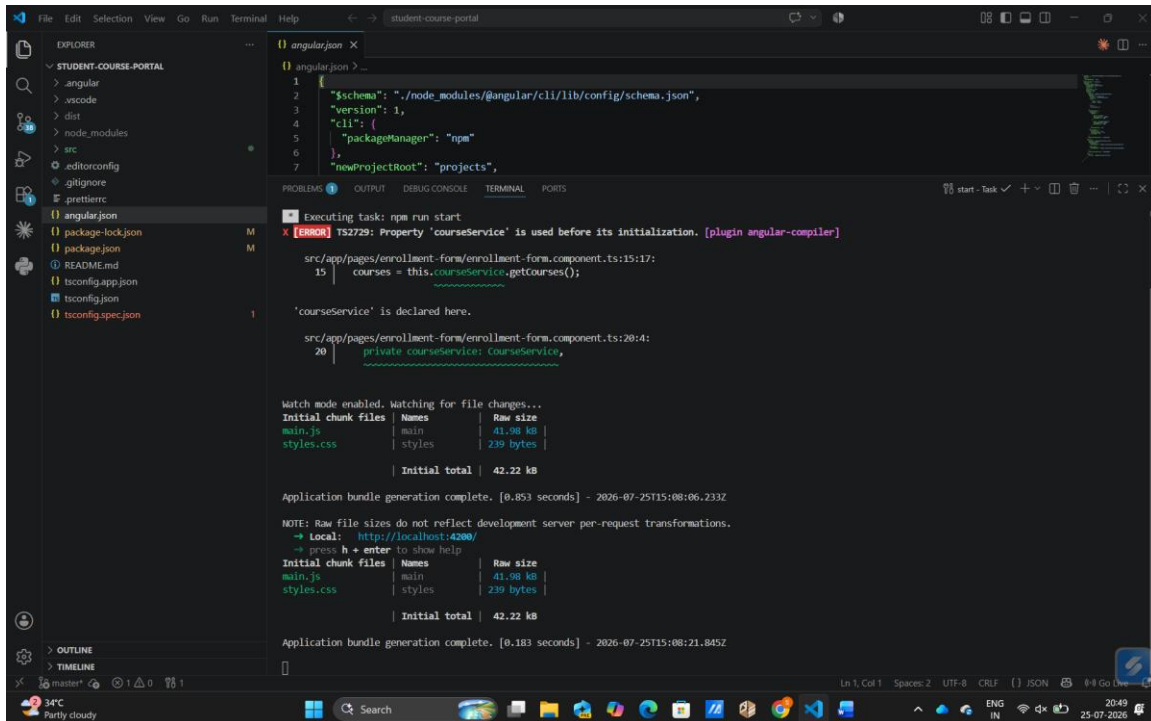
1. Implement canActivate in auth.guard.ts.
 - If isLoggedIn = true, allow navigation.
 - Otherwise, show alert and redirect to Home.
2. Apply guard to /profile route in app-routing.module.ts.

Task 3: Simulate Login/Logout

Goal: Demonstrate guard behavior.

Steps:

1. Add helper methods login() and logout() in auth.guard.ts.
2. Call login() manually in code to simulate authentication.
3. Test navigation again — Profile should now be accessible.



The screenshot shows a Visual Studio Code editor window for a project named 'student-course-portal'. The Explorer sidebar on the left shows the project structure with files like angular.json, package-lock.json, package.json, README.md, tsconfig-app.json, tsconfig.json, and tsconfig.spec.json. The main editor area displays the angular.json file, which is a JSON configuration for the Angular CLI. Below the editor, the PROBLEMS pane shows a TypeScript error: 'ERROR TS2729: Property 'courseService' is used before its initialization. [plugin angular-compiler]'. The error points to line 15 in the file src/app/pages/enrollment-form/enrollment-form.component.ts. The terminal pane at the bottom shows the output of the 'npm run start' command, indicating that the application bundle generation is complete and the development server is running at http://localhost:4200/. The terminal also displays file size information for the initial chunk files (main.js and styles.css) and the total initial bundle size (42.22 kB).

```
angular.json
1 {
2   "$schema": "./node_modules/@angular/cli/lib/config/schema.json",
3   "version": 1,
4   "cli": {
5     "packageManager": "npm"
6   },
7   "newProjectRoot": "projects",
8 }

src/app/pages/enrollment-form/enrollment-form.component.ts:15:17:
15 |   courses = this.courseService.getCourses();
   |                   ^^^^^^^^^^^^^^^^^^^^^^^^^
   | 'courseService' is declared here.

src/app/pages/enrollment-form/enrollment-form.component.ts:20:4:
20 |   private courseService: CourseService,
   |   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

Watch mode enabled. Watching for file changes...
Initial chunk files | Names | Raw size
main.js | main | 41.98 kB
styles.css | styles | 239 bytes
Initial total | 42.22 kB

Application bundle generation complete. [0.853 seconds] - 2026-07-25T15:08:06.233Z

NOTE: Raw file sizes do not reflect development server per-request transformations.
→ Local: http://localhost:4200/
→ press h + enter to show help

Initial chunk files | Names | Raw size
main.js | main | 41.98 kB
styles.css | styles | 239 bytes
Initial total | 42.22 kB

Application bundle generation complete. [0.183 seconds] - 2026-07-25T15:08:21.845Z
```

